

Constructing and Solving Equations — Paper C

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every equation on this paper is designed to come out exactly.
- The method is not printed on this paper. You are expected to know it.
- When a bracket is subtracted, the minus sign multiplies every term inside it. When a fraction wraps a whole side, multiplying by the denominator undoes it in one move.
- Several questions ask you to explain, prove or diagnose. A correct calculation is not by itself an explanation — say what the algebra shows.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – BRACKETS, AND FRACTIONS

1. Solve:

(a) $3(2x + 1) = 4x + 15$ [2]

(b) $5(x - 2) = 3(x + 2)$ [2]

2. Solve:

(a) $4(2x - 3) - 2(x - 5) = 22$ [2]

(b) $3(x + 4) - 2(x - 1) = 20$

[2]

3. Solve:

(a) $x \div 4 + 3 = 8$

[1]

(b) $(x - 3) \div 5 = 4$

[1]

(c) $2x \div 3 = 8$

[1]

(d) $(2x + 1) \div 3 = 5$

[1]

SECTION 2 – NEGATIVES, AND COMPARISONS

4. Solve:

(a) $7 - 2x = 1$

[2]

(b) $10 - 3x = 4x - 4$

[2]

5. Two gyms charge differently.

Gym A: a 300 rupee joining fee, then 40 rupees a month.

Gym B: a 100 rupee joining fee, then 60 rupees a month.

(a) Write an expression for the total cost at each gym after m months. [1]

(b) Form and solve an equation to find after how many months the two totals are equal. [2]

(c) Which gym is cheaper over 15 months, and by how much? [1]

SECTION 3 – REASONING IT OUT

6. Three angles lie on a straight line. They measure $(2x + 10)^\circ$, $(3x - 5)^\circ$ and $(x + 25)^\circ$.

(a) Form an equation in x . State the angle fact you have used. [1]

(b) Solve it. [2]

(c) Work out the three angles, and check that they are consistent with your angle fact. [2]

7. Three consecutive whole numbers add up to 72.

(a) Using n for the smallest, form an equation.

[1]

(b) Solve it and write down the three numbers.

[2]

(c) Explain why three consecutive whole numbers can never add up to 70.

[2]

End of paper. Check every answer has a unit or a form the question actually asked for.